Multi-Objective RLVR: Beyond Scalar Reward

Research Direction — General Framework

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RL Post-Training Compresses Everything Into One Number

Every major RL post-training paradigm reduces evaluation to a scalar:

Paradigm	**Reward**	**What gets compressed**
RLHF	Single RM score	Helpful + harmless + honest
DPO	Pairwise preference	Multi-dimensional quality
RLVR	Binary pass/fail	Correctness + efficiency + robustness
LLM-as-judge	Single score	Style + accuracy + reasoning

Practitioners care about many things: correctness, efficiency, robustness, reasoning quality, solution diversity, code style, edge case handling. All of it gets collapsed into one scalar before the policy ever sees it.

What if we stopped collapsing?

The Consequences of Scalar Reward

Scalar reward creates three systematic failure modes:

1. Mode collapse. The policy sharpens on one solution strategy that maximizes the scalar. Diversity is not rewarded, so it's destroyed. Pass@1 goes up, Pass@k goes down.

2. Objective neglect. Whichever dimension of quality contributes most to the scalar dominates. A solution that's correct but fragile, inefficient, or poorly reasoned gets the same $R = 1$ as a robust, elegant one.

3. Misaligned optimization. The policy optimizes for the measured objective (pass the test suite) at the expense of unmeasured objectives (handle unseen inputs, reason correctly, generalize). Goodhart's Law applied to verification.

These aren't bugs in specific RLVR algorithms — they're consequences of **the scalar**

The Proposal: Multi-Objective RLVR

Maintain a **vector reward** $\mathbf{R}(s, a) = [R_1, R_2, \ldots, R_K]$ and optimize the policy over all objectives simultaneously, without premature scalar collapse.

Three layers of the framework:

1. **Multi-objective verification** — decompose the verifier into multiple evaluation criteria
2. **Multi-dimensional reward generation** — produce dense signals along each axis
3. **Multi-objective policy optimization** — optimize GRPO over the vector, not the scalar

Each layer can be adopted independently. A practitioner can start with Level 1 (just run more checks) and progressively add Levels 2 and 3 as the infrastructure matures.

Level 1: Multi-Objective Verification

Instead of one test suite producing pass/fail, decompose verification into independent criteria:

Domain	Objective	Verification method
Code	Functional correctness	Test suite pass/fail
Code	Efficiency	Execution time within budget
Code	Robustness	Edge-case / fuzz test suite
Code	Readability	Linter + style checker score
Math	Answer correctness	Symbolic / numeric match
Math	Reasoning validity	Step-level verification
Math	Solution elegance	Length + complexity metric
Agents	Task completion	Success/fail
Agents	Efficiency	Number of steps / API calls
Agents	Safety	Constraint violation check

Level 2: Multi-Dimensional Dense Reward

Binary multi-objective verification is useful but still coarse. For richer signal, evaluate each axis densely:

Option A: Automated dense metrics

- Correctness: fraction of tests passed (k/n), or execution feedback analysis
- Efficiency: normalized execution time relative to known optimal
- Robustness: fraction of fuzz tests passed
- Each is continuous, cheap to compute, and objective

Option B: LLM-based dense evaluation

- A judge model (external or self) evaluates along each axis via different prompts
- Produces structured judgment per axis (e.g., efficiency: optimal / acceptable / wasteful)

Level 3: Multi-Objective Policy Optimization

The hard part. GRPO computes scalar advantages: $\hat{A}_i = (r_i - \mu)/\sigma$

With vector reward $\mathbf{R} \in \mathbb{R}^K$, we need a new advantage formulation.

Four candidate approaches:

Method	Mechanism	Pro	Con
Scalarization	$R = \sum \alpha_k R_k$	Simple	Collapses to one point
Pareto GRPO	Keep Pareto-improving	Principled diversity	Complex, no prior work
Lexicographic	Priority ordering	Preserves primary obj	Rigid hierarchy
Conditional	$\alpha(s)$ adapts per problem	Automatic curriculum	Needs difficulty signal

These are not mutually exclusive — lexicographic ordering can use Pareto within each priority tier.

Scalarization: Simple but Lossy

$$R_{\text{total}}(s,a) = \sum_{k=1}^{K} \alpha_k \cdot R_k(s,a) \qquad \hat{A}_i = \frac{R_{\text{total},i} - \mu}{\sigma}$$

Standard GRPO, just with a weighted sum reward. Each weight vector $\boldsymbol{\alpha}$ selects one point on the Pareto frontier.

Limitation: The policy converges to whichever solution maximizes the weighted sum. Diversity is determined by $\boldsymbol{\alpha}$, not by the training process.

Interesting connection: Running K different scalarizations (different $\boldsymbol{\alpha}$ vectors) is equivalent to population-based RLVR with different reward preferences. Each policy finds a different Pareto point. Cross-pollinating the results recovers the full Pareto frontier.

Pareto GRPO: Principled but Novel

Define per-objective advantages: $\hat{A}_i^{(k)} = (R_{k,i} - \mu_k)/\sigma_k$ for each axis k.

A rollout is **Pareto-improving** if $\hat{A}_i^{(k)} > 0$ for at least one k and $\hat{A}_i^{(k)} \geq 0$ for all k.

Pareto GRPO update:

$$\nabla_\theta \mathcal{L} = -\mathbb{E}\left[\sum_{k=1}^{K} \beta_k(\hat{\mathbf{A}}_i) \cdot \hat{A}_i^{(k)} \cdot \nabla_\theta \log \pi_\theta(a_i \mid s)\right]$$

where $\beta_k(\hat{\mathbf{A}}_i)$ upweights axes on which rollout i is Pareto-improving.

Why this preserves diversity: A rollout that's mediocre on correctness but excellent on efficiency still gets positive gradient — it's Pareto-improving on the efficiency axis. Standard scalar GRPO would discard it. Pareto GRPO retains it, maintaining pressure toward diverse solution strategies.

Lexicographic GRPO: Practical and Safe

Impose a priority ordering on objectives: correctness $\succ$ robustness $\succ$ efficiency $\succ$ diversity.

Algorithm:

1. Compute advantages using primary objective (correctness)
2. Among rollouts with equivalent primary advantage (e.g., all correct), re-rank by secondary objective (robustness)
3. Among rollouts with equivalent secondary, re-rank by tertiary, etc.

Why this is safe: Correctness is never sacrificed for secondary objectives. The hierarchy guarantees Pass@1 is never worse than standard GRPO. But among correct solutions, the policy now differentiates — it prefers robust, efficient, diverse correct solutions over fragile, wasteful, repetitive ones.

Conditional Objectives: Automatic Curriculum

The optimal weighting depends on the problem:

Problem regime	**Pass rate p**	**Optimal α**
Easy ($p > 0.8$)	Most rollouts correct	Emphasize efficiency, elegance
Medium ($0.2 < p < 0.8$)	Mixed pass/fail	Balance correctness + diversity
Hard ($p < 0.2$)	Almost all fail	Maximize correctness only
Solved ($p \approx 1.0$)	All correct	Pure diversity / robustness

$$\boldsymbol{\alpha}(s) = f(p(s)) \qquad \text{where } p(s) \text{ is estimated pass rate for prompt } s$$

This is multi-objective as **automatic curriculum**: hard problems get pure correctness pressure, easy problems get pressure toward diversity and efficiency, medium problems get balanced multi-objective optimization. No manual curriculum design needed — the objective vector adapts to the policy's current capability per problem.

Theoretical Properties

Proposition (Pareto diversity): Under Pareto GRPO with $K \geq 2$ objectives, the set of rollouts receiving positive gradient is strictly larger than under any scalarization. This guarantees the policy maintains probability mass on a broader set of solution strategies.

Proposition (lexicographic safety): Under lexicographic GRPO with correctness as the primary objective, $\mathrm{Pass@1}_{\text{lexi}} \geq \mathrm{Pass@1}_{\text{scalar}}$. Secondary objectives never degrade the primary.

Proposition (conditional as optimal scalarization): If $\boldsymbol{\alpha}(s)$ adapts to the problem's difficulty, conditional GRPO recovers the optimal scalarization for each prompt, outperforming any fixed $\boldsymbol{\alpha}$.

These are provable under standard policy gradient assumptions. Each constitutes a theoretical contribution.

What This Framework Unifies

Several existing ideas are special cases of multi-objective RLVR:

Existing method	**Multi-objective interpretation**
Standard RLVR	$K = 1$ (correctness only)
RLVR + entropy bonus	2 objectives: correctness + entropy
DIVER / DSDR	2 objectives: correctness + diversity
Pass@k optimization	2 objectives: correctness + coverage
DPH-RL (forward-KL)	Implicit: correctness + base model coverage
Reward shaping (dense)	Richer R_1 but still 1 objective

Each is a hand-crafted scalarization of two objectives. Our framework generalizes: define the objectives explicitly, optimize over the vector directly, let the Pareto structure determine diversity rather than hand-tuning entropy coefficients.

Experimental Plan

Phase 1: Multi-objective verification (2 weeks)

- Decompose code verification into 4 axes: correctness, efficiency, robustness (fuzz tests), style
- Run standard GRPO with each axis as a binary signal
- Measure axis independence (correlation matrix). Go/no-go: $<$0.8 pairwise correlation

Phase 2: Lexicographic GRPO (4 weeks)

- Implement lexicographic advantage computation (correctness $\succ$ robustness $\succ$ efficiency)
- Compare: standard GRPO, scalarized GRPO, lexicographic GRPO
- Metrics: Pass@1 (must not degrade), Pass@k, solution diversity, efficiency distribution

Key Metrics: What Multi-Objective RLVR Should Improve

Standard metrics only measure scalar performance. We need a multi-objective evaluation suite:

Metric	What it measures
Pass@1	Single-attempt correctness (must not degrade)
Pass@k (k=32, 64, 256)	Solution coverage / diversity
Efficiency@1	Avg execution time of Pass@1 solutions
Robustness@1	Fuzz test pass rate of Pass@1 solutions
Solution family count	Number of distinct strategy clusters
Pareto volume	Hypervolume of the Pareto frontier
Conditional Pass@1	Pass@1 stratified by difficulty tier

A method that improves Pass@1 while degrading Robustness@1 and Pass@k has failed the multi-objective criterion, even though it would look good on standard benchmarks.

Risks and Mitigations

Risk	Severity	Mitigation
Pass@1 degrades (practitioners won't adopt)	**High**	Lexicographic: primary obj is correctness
Objectives aren't independent	High	Phase 1 measures correlation; drop redundant axes
Pareto GRPO is intractable	Medium	Start with lexicographic (simpler)
No benchmark measures multi-objective	Medium	Define evaluation suite; release as contribution
Overhead of multi-axis verification	Low	Most checks are cheap (linter, timer, fuzz)
Framework is too general (reviewer: "what's the	Medium	Focus on lexicographic GRPO as the concrete

Honest Assessment

Why this could be a strong paper:

- General framework that unifies existing ad-hoc diversity methods
- Lexicographic GRPO is simple, safe (never degrades Pass@1), and novel
- Theoretical properties are provable (Pareto diversity, lexicographic safety)
- Addresses the pass@k degradation problem by design, not by heuristic
- Defines a new evaluation paradigm (multi-objective metrics for RLVR)

Why it might not work:

- Multi-objective RL for LLMs is unexplored territory — infrastructure doesn't exist
- Pareto GRPO formulation needs theoretical work to be rigorous
- Reviewers may see it as "just adding more reward terms" (need strong framing)